Shri Siddhivinayak mahila Mahavidyalaya

PROJECT TOPIC

MoodFlix – Movie Recommendation System Based on Emotions

TY BSC COMPUTER SCIENCE VI

✦ Student Name (Group)

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✦ Programming Language / Platform (OS)

Programming Language : Python

Frontend Technology : HTML, CSS, Java

Backend Framework : Flask

Database : SQLite / CSV Dataset

Machine Learning Libraries : OpenCV, TensorFlow / Keras, NumPy

Platform / OS : Linux

1.Introduction

MoodFlix is a smart movie recommendation system the suggests movies to users based on their emotional state.
The system uses facial emotion recognition technology to detect the user's current mood through a webcam and recommends suitable movies accordingly.

In today's digital era, users often spend a lot of time searching for movies that match their mood. MoodFlix simplifies this process by automatically analyzing emotions such as **Happy, Sad, Angry, Surprise, and Neutral**, and recommending movies that best align with the detected emotion. The project combines **machine learning, computer vision, and web technologies** to create an interactive and user-friendly recommendation system.

Moreover, MoodFlix aims to enhance the overall entertainment experience by providing personalized movie suggestions without requiring manual input from the user. By using real-time emotion detection, the system adapts to the user's mood instantly, making movie selection faster and more accurate. This approach not only improves user satisfaction but also highlights the practical application of artificial intelligence in daily life. MoodFlix can be useful for individuals who find it difficult to decide what to watch or want recommendations that truly reflect how they feel at a given moment. The system also demonstrates how emerging technologies such as artificial intelligence, machine learning, and computer vision can be effectively integrated to solve real-world problems in the field of digital entertainment.

2.Motivation

With the rapid growth of online streaming platforms, users are often overwhelmed by the vast amount of available movie content. Selecting a movie that matches one's mood has become time-consuming and confusing. Existing recommendation systems primarily depend on genres, ratings, or previous viewing history and fail to consider the user's emotional state.

This gap motivated the development of MoodFlix, which automatically detects user emotions through facial expressions and provides personalized movie recommendations, enhancing the overall entertainment experience.

In addition to this, understanding a user's emotional state can significantly improve the relevance of movie recommendations. Emotions play an important role in decision-making, especially in entertainment choices. By incorporating facial emotion recognition, MoodFlix addresses the limitations of traditional recommendation systems and introduces a more human-centric approach. This motivation also reflects the growing need for intelligent systems that can adapt to user behavior in real time. The project encourages the use of advanced technologies such as artificial intelligence and machine learning to build smarter applications that enhance user engagement and satisfaction in digital platforms.

3.Problem Statement

Users frequently face difficulty in choosing movies that align with their emotional state, leading to dissatisfaction and wasted time. Traditional recommendation systems do not offer real-time emotion-based suggestions and require manual input from users.

There is a need for an intelligent system that can automatically detect emotions in real time and recommend suitable movies accordingly.

MoodFlix addresses this problem by utilizing facial emotion recognition to provide accurate and emotion-specific movie recommendations without user intervention.

4.Purpose / Objectives / Goals

The primary objectives of the MoodFlix system are:

- To detect the user’s emotional state using facial expression analysis
- To provide accurate movie recommendations based on detected emotions
- To minimize the time required to search for mood-appropriate movies
- To develop an interactive and user-friendly web interface
- To efficiently integrate machine learning and web technologies

5.Literature Survey

Several movie recommendation systems have been proposed using content-based filtering, collaborative filtering, and hybrid recommendation techniques. Recent research highlights the effectiveness of emotion-aware recommendation systems in improving user satisfaction. Facial emotion recognition using OpenCV and Convolutional Neural Networks (CNNs) has demonstrated high accuracy in identifying human emotions such as happiness, sadness, anger, and surprise.

The MoodFlix system builds upon these studies by combining facial emotion detection with a web-based movie recommendation framework to deliver enhanced personalization.

6,Project Scope

- The scope of the MoodFlix project includes
- Real-time facial emotion detection using a webcam
- Emotion-based movie recommendation mechanism
- Web application development using HTML, CSS, and Flask
- Use of an emotion-categorized movie dataset
- Suitable for personal use and academic purposes
- Potential application in entertainment and streaming platforms

7.Project Limitations

- Requires a functional webcam for emotion detection
- Performance depends on lighting conditions and camera quality
- Limited to a fixed set of predefined emotions
- Recommendation quality depends on dataset size and accuracy
- Not scalable for large commercial platforms without further optimization

✦ Scope of the System

- The system detects user emotions using real-time facial expression analysis.
- It recommends movies based on the detected emotional state.
- The system provides a simple and interactive web-based interface.
- It reduces the time spent searching for suitable movies.
- MoodFlix can be used for entertainment platforms and personal use.
- The system is suitable for small to medium-scale applications.

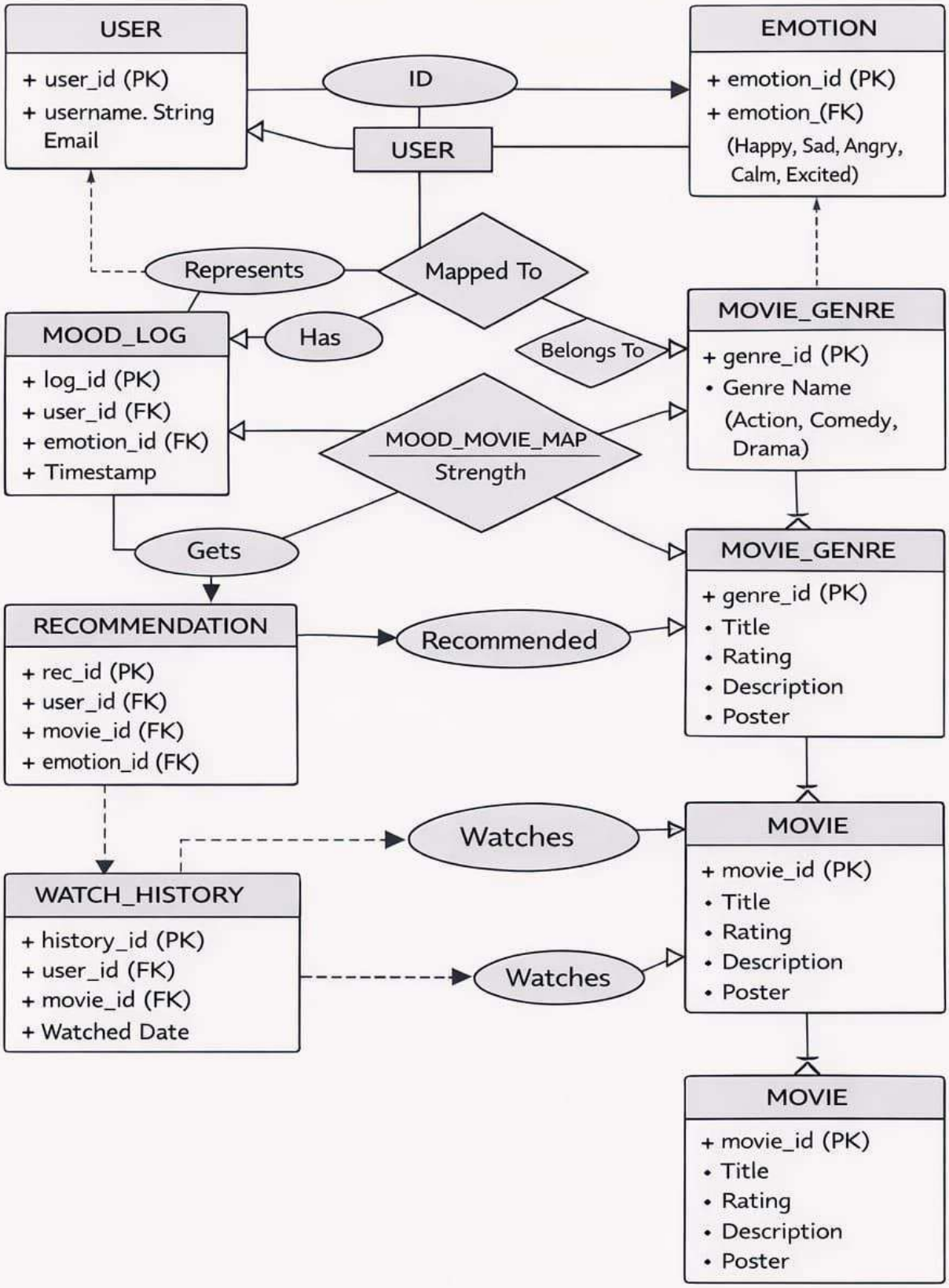
✨ New Features Implemented by Student

- Real-time facial emotion detection using webcam
- Emotion-based movie recommendation system
- Categorized movie dataset based on emotions
- User-friendly web interface
- Automated recommendation without manual input

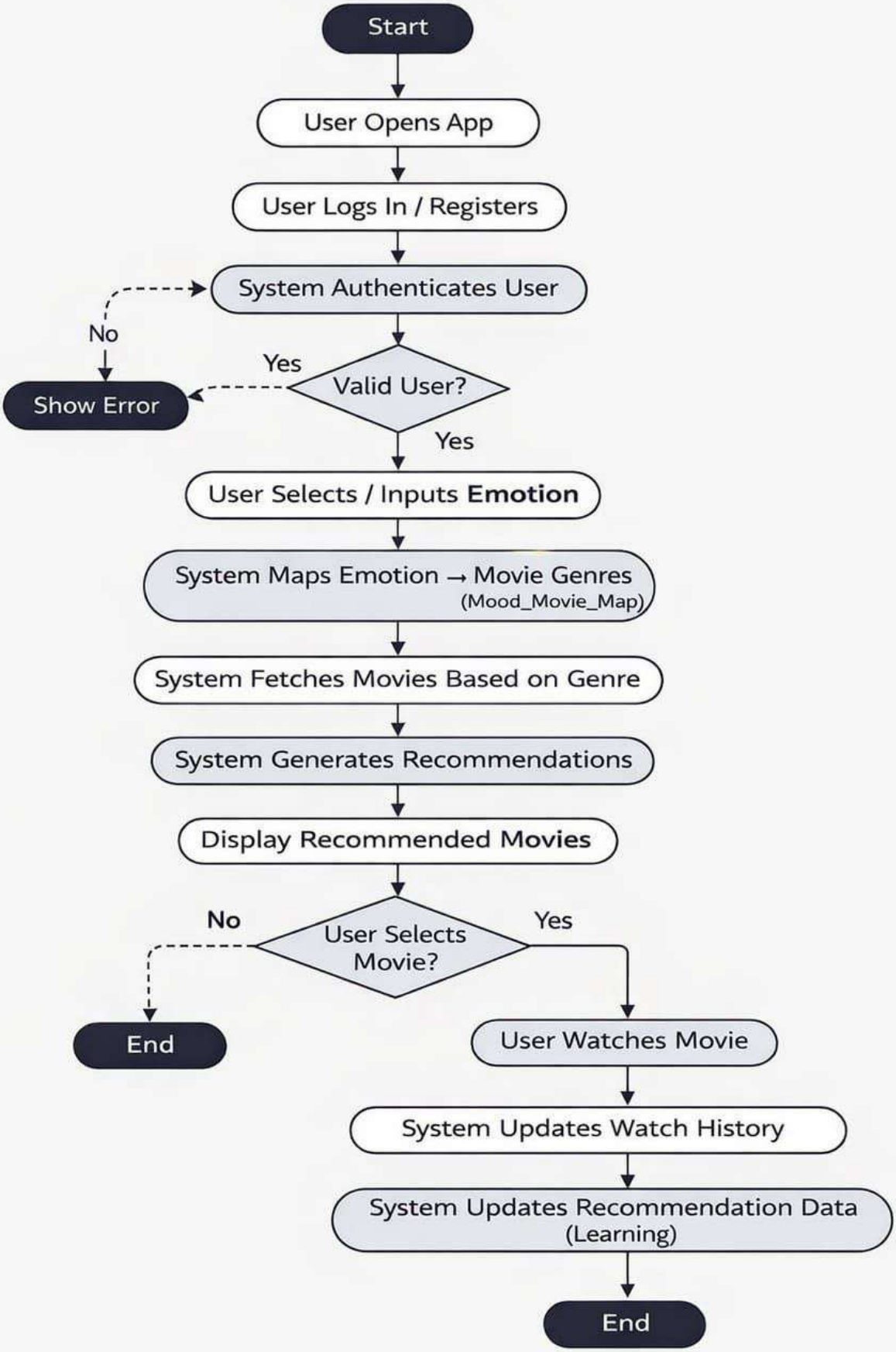
Acadmic Year : 2025-2026

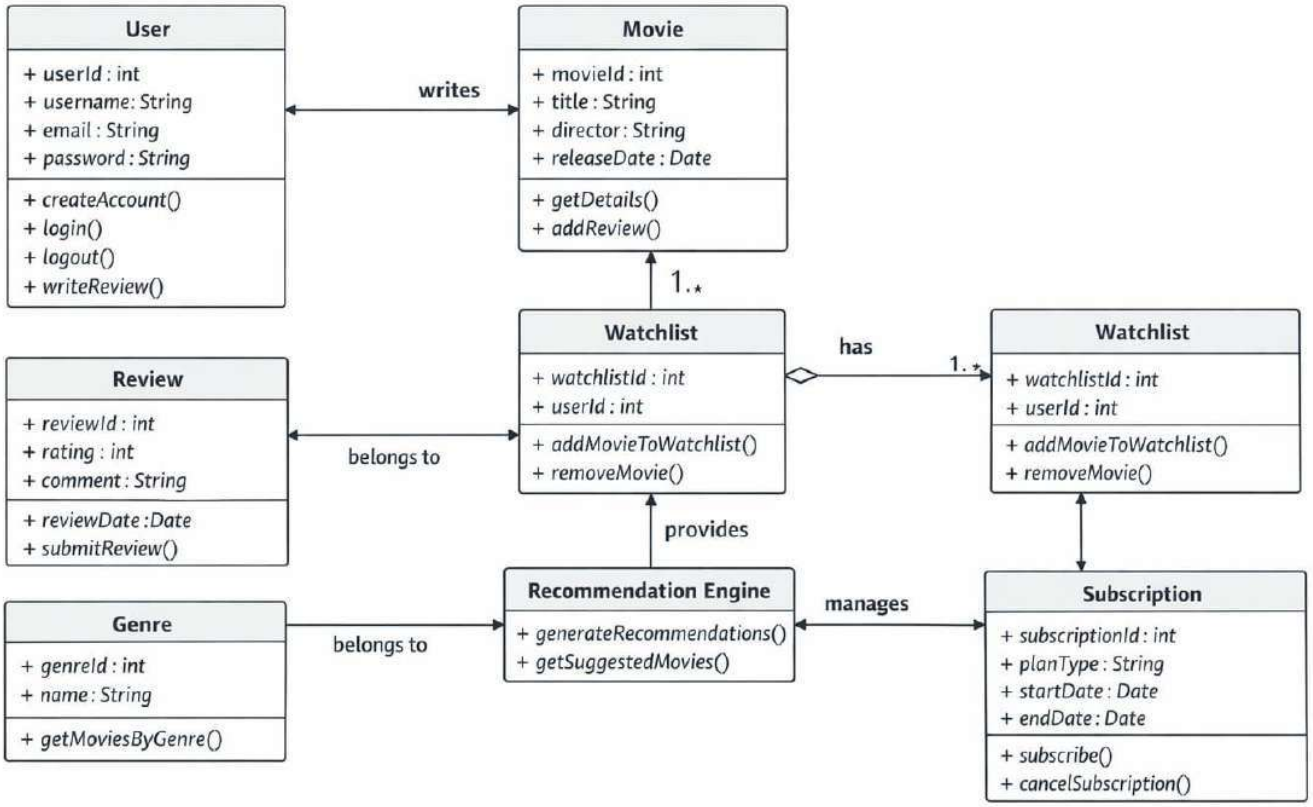
Department : Computer Science

MoodFlix: Movie Recommendation System Based on User Emotions



MoodFlix: Movie Recommendations System Based on User Emotions

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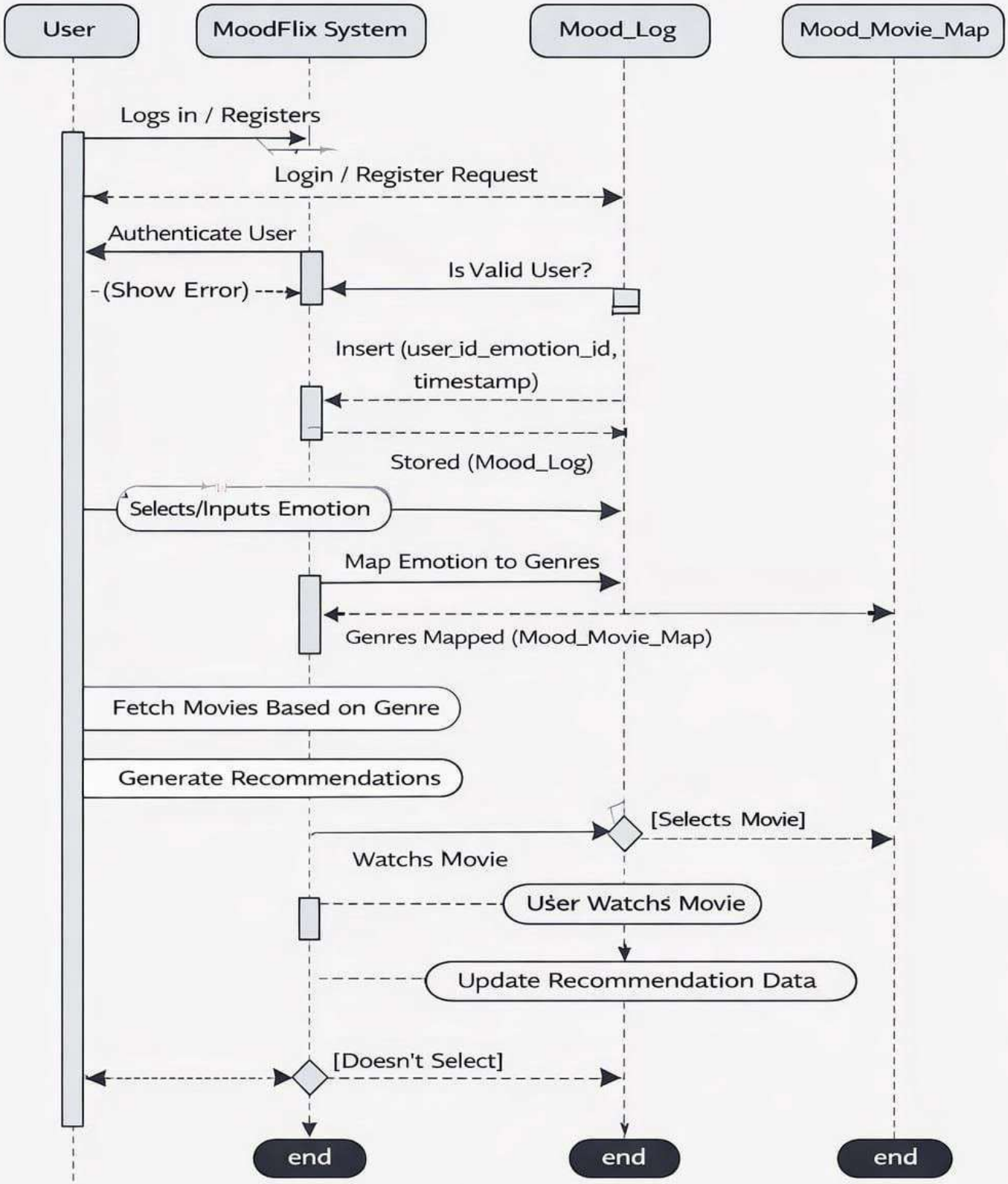
Sequence Diagram – MoodFlix: Movie Recommendation System Based on User Emotions



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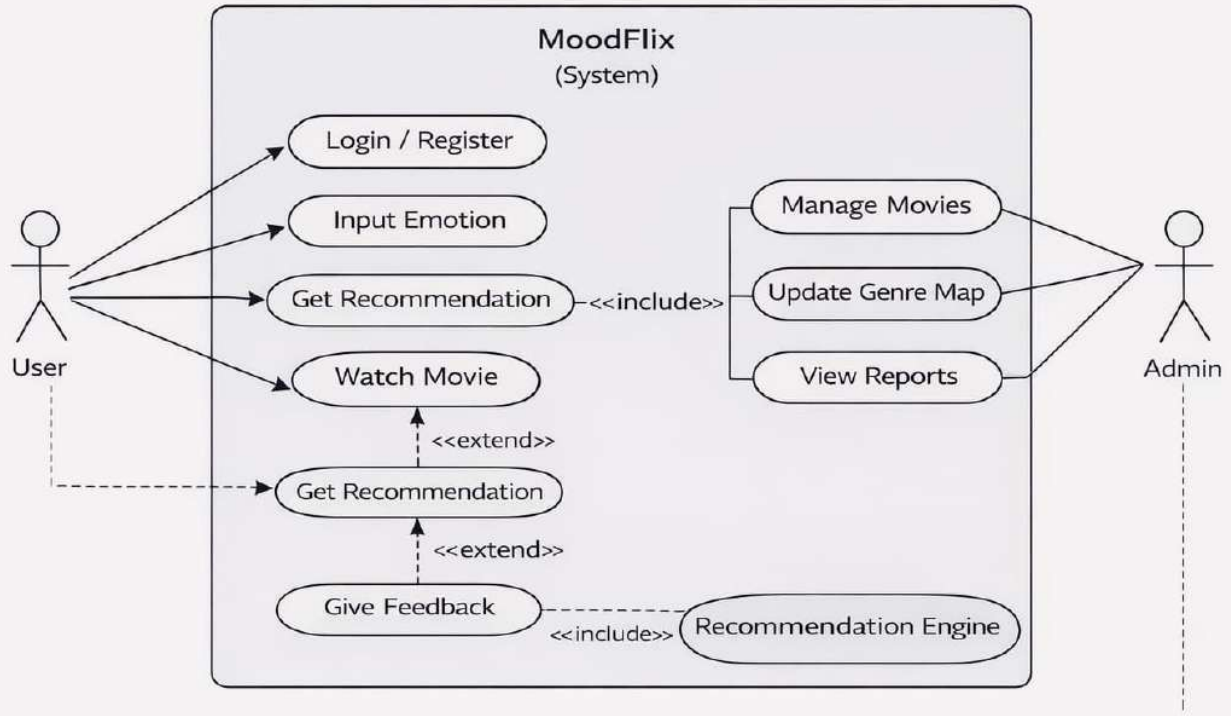

1.Home page:

Test Case Id	Test Case Name	Step to execute	Input Data	Expected output	Actual output	status	Description
TC001	Home page button	Navigate to Home page	N/A	Redirect to home page	Redirect to home page	pass	Ensure that home button redirected home page
TC002	Start Recommendation Button	Proceed to mood selection page	N/A	Access to mood selection page	Access to mood selection page	Pass	This button is to get access of mood selection
TC003	About us	Navigate to about details	N/A	Display project details	Display project details	Pass	Display project details to user
TC004	Features	Navigate features of website	N/A	Display the features of website	Display the features of website	pass	Show the features of website to user
TC005	Signup Button Visibility	Navigate to the index page	N/A	Signup button is visible	Signup button displayed	Pass	Check if the Signup button is displayed
TC006	Login Button Visibility	Navigate to the index page	N/A	Login button is visible	Login button is displayed	Pass	Ensure that the Login button is displayed

2.Login page:

Test Case Id	Test Case Name	Step to execute	Input Data	Expected output	Actual output	status	Description
TC001	Email button	Get email address	Email id	Get email id	Get email id	pass	Get email id in system
TC002	Password button	Get password	password	Display password	display password	pass	Get password to verify the user
TC003	Login button	Click login button	Valid credentials	Login successful	Login successful	Pass	Ensure successful login with valid data
TC004	Invalid login	Enter wrong credentials	Wrong ema il/password	Display error message	Display error message	Fail	Check invalid login handling
TC005	Empty fields	Leave fields blank and click login	N/A	Validation message should appear	Validation message shown	Fail	Ensure all login fields are mandatory

3.Signup page:

Test Case Id	Test Case Name	Step to execute	Input Data	Expected output	Actual output	status	Description
TC001	Full Name field	Enter full name	Rutuja	Accept full name	Accept full name	Pass	Verify full name input
TC002	Email field	Enter email address	user@gmail.com	Accept email id	Accept email id	Pass	Verify signup email
TC003	Password field	Enter password	123456	Accept password	Accept password	Pass	Verify signup password
TC004	Signup button	Click signup button	Valid details	Account should be created	Account created	Pass	Check signup process
TC005	Duplicate email	Enter already registered email	Existing email	Error should appear	Error displayed	Fail	Prevent duplicate account
TC006	Empty signup fields	Leave fields blank	N/A	Validation message should appear	Validation message shown	Fail	Ensure signup fields are required

4.Mood Selection page:

Test Case Id	Test Case Name	Step to execute	Input Data	Expected output	Actual output	status	Description
TC001	Mood options display	Open mood selection page	N/A	Mood options should be visible	Mood options displayed	Pass	Ensures moods are shown
TC002	Happy mood	Select Happy mood	Happy	Mood should be selected	Happy selected	Pass	Checks happy mood input
TC003	Sad mood	Select Sad mood	Sad	Mood should be selected	Sad selected	Pass	Checks sad mood input
TC004	Angry mood	Select Angry mood	Angry	Mood should be selected	Angry selected	Pass	Checks angry mood input
TC005	Relaxed mood	Select Relaxed mood	Relaxed	Mood should be selected	Relaxed selected	Pass	Checks relaxed mood input
TC006	Excited mood	Select Excited mood	Excited	Mood should be selected	Excited selected	Pass	Checks excited mood input
TC007	No mood selected	Click continue without mood	N/A	Error message should appear	Error displayed	Fail	Prevents empty mood submission

5.Recommendation page:

Test Case Id	Test Case Name	Step to execute	Input Data	Expected output	Actual output	status	Description
TC001	Recommendation Generation	Select mood and continue	Happy	Movie list should be generated	Movie list displayed	Pass	Checks recommendation system
TC002	Movie Title Display	Open recommendation page	N/A	Movie titles should be shown	Titles displayed	Pass	Ensures titles are visible
TC003	Genre Display	Open recommendation page	N/A	Genre should be shown	Genre displayed	Pass	Verifies movie genre visibility
TC004	Description Display	Open recommendation page	N/A	Movie description should be shown	Description displayed	Pass	Checks movie description
TC005	Relevant Recommendations	Select Sad mood	Sad	Sad / emotional movies should appear	Correct movies displayed	Pass	Ensures recommendations match mood
TC006	Page Load Speed	Open recommendation page	N/A	Page should load quickly	Page loaded fast	Pass	Verifies performance